
Question #: 1

A 20-year-old man comes to the physician because of a 2-month history of lack of sweating and flushing. Neurological examination shows damage to postganglionic sympathetic cell bodies. Which of the following structures is most likely damaged?

A. Brainstem
✓B. Sympathetic chain
C. Dorsal root ganglia
D. Lateral gray horn
E. Ventral horn

Rationale: Postganglionic sympathetic cell body in found in sympathetic chain ganglia.

Question #: 2

A 20-year-old man comes to the emergency department because of a 2-hour history of a painful laceration to the skin of his upper limb. Which of the following fibers most likley transmits the pain sensation to the central nervous system?

A. Visceral afferent
B. Somatic efferent
✓C. Somatic afferent
D. Visceral efferent
E. Autonomic

Rationale: Somatic fibers carry information too and from body surface and afferent fiber will carry information to central nervous system.

Question #: 3

A 5-year-old girl is brought to the physician by her mother because of a 1-year history of failure to thrive. Laboratory studies show serum folate levels and blood ammonia levels within the reference ranges. A complete blood picture shows megaloblastic anemia. Urinalysis shows elevated orotic acid levels. Targeted gene mutation analysis shows an inherited deficiency of orotate phosphoribosyltransferase. Which of the following pathways is most likely impaired in this patient?

✓A. De novo synthesis of pyrimidine nucleotides

B. De novo synthesis of purine nucleotides
C. Formation of uric acid
D. Catabolism of pyrimidines
E. Salvage of purine nucleotides

Rationale: The child in the vignette has orotic aciduria due to deficiency of orotate phosphoribosyltransferase (OPRT). This is an enzyme of the pyrimidine biosynthesis pathway. Orotic aciduria due to pyrimidine biosynthesis defect results in reduced synthesis of the pyrimidine nucleotides, uridine, cytidine and thymidine. This results in a delay in RBC maturation and megaloblastic anemia. Dietary uridine supplementation provides the pyrimidine precursor for cytidine and thymidine synthesis.

Question #: 4

A 40-year-old man comes to the physician because of a 2-week history of itching especially after a warm shower. After a thorough clinical assessment, he is diagnosed with a hematological malignancy called polycythemia vera. He is prescribed hydroxyurea. Which of the following processes is most likely inhibited by this drug?

A. AMP synthesis from IMP
✓B. Deoxyribonucleotide synthesis
C. Uric acid formation from purine nucleotides
D. Thymidylate synthesis from dUMP
E. Reduction of dihydrofolate to tetrahydrofolate
F. GMP formation from IMP

Rationale: Hydroxyurea inhibits the enzyme ribonucleotide reductase which is required for the formation of deoxyribonucleotides from ribonucleotides. Review drugs that act as inhibitors of the other steps in nucleotide metabolism.

Allopurinol inhibits uric acid formation from purine nucleotides.
Thymidylate synthase is inhibited by 5-fluoro uracil.
Dihydrofolate reductase in eukaryotes is inhibited by methotrexate. Bacterial dihydrofolate reductase inhibitor is trimethoprim.
Mycophenolic acid inhibits IMP dehydrogenase which converts IMP to GMP.

Question #: 5

A 40-year-old man comes to the physician because of a 1-day history of severe pain and redness in the right great toe. He is prescribed analgesics and a xanthine oxidase inhibitor for long-term management of arthritis. Which of the following pharmacotherapy is most likely prescribed to the patient?

A. Mycophenolic acid
✓B. Allopurinol
C. 5-Fluorouracil
D. Quinolones
E. Methotrexate
F. Sulfonamide

Rationale: The clinical presentation is gouty arthritis. Allopurinol inhibits the formation of uric acid from xanthine and hypoxanthine and is used in patients with gout.
Mycophenolic acid inhibits GMP synthesis from IMP.
5-Fluorouracil is an inhibitor of thymidylate synthase. Quinolones (ciprofloxacin) are antibacterial agents that are DNA gyrase inhibitors. Methotrexate inhibits dihydrofolate reductase. Sulfonamides inhibit folic acid synthesis in bacteria and is used an an antibacterial agent.

Question #: 6

A 1-month-old boy is brought to the physician by his parents for a follow-up evaluation. He has had repeated respiratory and gastrointestinal infections since birth. Laboratory studies show a severe reduction in T- and B-cell function. Cytogenomic microarray analysis shows no abnormalities. Neutrophil respiratory burst activity is present on phagocytosis. Which of the following is the most likely diagnosis?

✓A. Adenosine deaminase deficiency
B. DiGeorge syndrome
C. Orotic aciduria due to UMP synthase deficiency
D. Chronic granulomatous disease
E. Lesch Nyhan syndrome

Rationale: The vignette describes a patient with SCID (Severe combined immunodeficiency) due to adenosine deaminase deficiency. Note the reduction in T and B cells and the occurrence of repeated infections. Accumulation of dATP inhibits ribonucleotide reductase and inhibits lymphocyte proliferation in patients with ADA deficiency. ADA catalyzes the conversion of adenosine to inosine in purine nucleotide degradation.

DiGeorge syndrome (22q microdeletion syndrome) - is characterized by thymic aplasia and other associated findings. Microarray analysis would show a deletion in chromosome 22.

Orotic aciduria is due to a deficiency of pyrimidine biosynthesis and these patients have megaloblastic anemia and normal blood ammonia levels. They do NOT have immunodeficiency.

Chronic granulomatous disease is characterized by a deficiency of NADPH oxidase. Patients have repeated bacterial and fungal infections and neutrophil respiratory burst activity is absent.

Lesch-Nyhan syndrome is an X-linked inherited disorder due to a deficiency of HGPRT (hypoxanthine-guanine phosphoribosyl transferase). Characterized by intellectual disability, self-mutilation, and hyperuricemia.

Question #: 7

A 5-year-old boy is brought to the physician by his parents because of a 2-month history of biting his fingers and lips. There is a delay in the developmental milestones. Laboratory studies show serum uric acid is 9.5mg/dL (3-8.2mg/dL). Laboratory studies show very low activity of an enzyme in intermediary metabolism. Which of the following enzyme-catalyzed reactions is most likely impaired in this patient?

A. Conversion of xanthine to uric acid
B. Formation of PRPP from ribose 5-phosphate
C. Formation of deoxyribonucleotides from ribonucleotides
D. Conversion of adenosine to inosine
✓E. Conversion of hypoxanthine to inosine monophosphate (IMP)

Rationale: The vignette describes a patient with the X-linked recessive disorder, Lesch-Nyhan syndrome. Note the developmental delay, intellectual disability, self-mutilation and hyperuricemia. The enzyme deficient in the patient is HGPRT (Hypoxanthine-guanine phosphoribosyl transferase) which converts hypoxanthine (free purine base) to Inosine monophosphate (IMP; Purine nucleotide) and guanine to GMP.

Conversion of xanthine to uric acid - Enzyme is Xanthine oxidase. Inhibited by allopurinol;

Formation of PRPP from ribose 5-phosphate - Catalyzed by PRPP synthetase;

Formation of deoxyribonucleotides from ribonucleotides - Catalyzed by ribonucleotide reductase;

Conversion of adenosine to inosine - Catalyzed by Adenosine deaminase (ADA). ADA deficiency results in autosomal recessive SCID; Characterized by increased susceptibility to infections.

Conversion of guanine to Guanosine monophosphate (GMP) - Catalyzed by HGPRT

Question #: 8

A 25-year-old woman comes to the clinic because of a 1-week history of abdominal pain and loose stools. Physical examination shows tenderness in the left lower quadrant of the abdomen. A complete blood count and liver function tests are within normal limits. Endoscopic examination confirms a diagnosis of Crohn's disease. Treatment is initiated. Two weeks later, her absolute neutrophil count (ANC) decreases from 5140/μL to 1010/μL, and her white blood cell count (WBC) decreases from 6270/μL to 2810/μL, prompting a reduction in the dosage of her medication. Genetic analysis shows the patient is homozygous for a mutation in the TPMT gene. Which of the following therapeutic agents is the patient most likely on?

A. Methotrexate
✓B. Azathioprine
C. 5-fluorouracil
D. Mycophenolic acid
E. Hydroxyurea
F. Trimethoprim

Rationale: Mercaptopurines (azathioprine) are inhibitors of purine biosynthesis. They are metabolized by TPMT to inactive compounds. Low activity of TPMT results in toxicity of these drugs: Anemia and bone marrow suppression leading to low hemoglobin levels and low WBC counts. In these individuals, the dose of the drug has to be lowered to minimize toxicity.
Methotrexate: Eukaryotic Dihydrofolate reductase inhibitor and inhibits purine and thymidine synthesis
5-fluorouracil inhibits thymidylate synthase (pyrimidine synthesis)
Mycophenolic acid inhibits the synthesis of GMP from IMP.
Hydroxyurea inhibits ribonucleotide reductase and the formation of deoxyribonucleotides.
Trimethoprim inhibits prokaryotic dihydrofolate reductase and is used as an antibacterial agent.

Question #: 9

A 19-month-old boy is brought to the physician by his parents for a follow-up examination. He has a 2-month history of anemia and has been prescribed oral vitamin B12, iron, vitamin B6, and folic acid supplements with no improvement in hematological parameters. Physical examination shows pallor, poorly formed fingernails, and sparse hair. A blood sample is drawn for routine diagnostics. Laboratory studies show megaloblastic anemia. Urinalysis shows elevated orotic acid. Administration of which of the following metabolites will most likely improve this patient's anemia?

✓A. Uridine
B. Deoxyadenosine triphosphate (dATP)
C. Inosine
D. Glutamine
E. Tetrahydrofolate

Rationale: The patient has orotic aciduria due to a deficiency of UMP synthase (OMP decarboxylase/ OPRT).
Megaloblastic, macrocytic anemia is due to reduced synthesis of pyrimidine nucleotides (Uridine, cytidine and thymidine nucleotides for DNA synthesis).
Uridine is the common precursor of UMP, CTP, and dTTP.
CPS-II is the regulated enzyme of pyrimidine nucleotide biosynthesis. It is allosterically inhibited by UTP; Uridine administration will inhibit excessive orotic acid formation by feedback inhibition of CPS-II.
PRPP is an activator of purine and pyrimidine nucleotide synthesis.
Glutamine is a substrate of CPS-II.
dATP is an inhibitor of ribonucleotide reductase.
Inosine (contains hypoxanthine linked to ribose; Purine nucleoside) is formed from adenosine by adenosine deaminase.

Question #: 10

A 32-year-old man comes to the physician because of a 4-week history of recurrent episodes of nervousness and palpitations. He says he develops these episodes when he is under stress and each episode lasts for about 15-20 minutes. His pulse is 110/min and blood pressure is 136/84 mm Hg. The physical examination shows anxiety, sweating, fine tremors of hands and dilated pupils. He is diagnosed to have stimulation of the sympathetic nervous system under stress. Which of the following best explains the effects in the target tissues when this division of the nervous system is active in the patient?

A. The chemical structure of the neurotransmitter
✓B. The properties of the postsynaptic receptor
C. The rate of uptake of neurotransmitter
D. Amount of calcium released from presynaptic axons
E. The velocity of axoplasmic transport in the presynaptic neuron

Rationale: Even though many neurotransmitters are generally classified as inhibitory or excitatory, the exact response in post-synaptic cells depends on which specific receptor sub-types mediate this response. Stimulation of sympathetic nervous system produces both excitatory and inhibitory effects depending on the postsynaptic receptors in the target tissues.

Question #: 11

A 34-year-old man comes to the physician because of a 1-week history of difficulty breathing through the nose while sleeping. He says he feels a kind of block in his nose when he goes to supine position. His vital signs are within normal limits. Physical examination shows redness and congestion in the nasal mucosa. He is diagnosed with nasal allergy. He is prescribed a nasal drop that contains an adrenergic receptor agonist, causing vasoconstriction in the nasal blood vessels. Which of the following adrenergic receptor types is the most likely target of this drug?

✓A. Alpha-1
B. Alpha-2
C. Beta-1
D. Beta-2
E. Beta-3

Rationale: Alpha-1 adrenergic receptor agonists stimulate vasoconstriction in the blood vessels and reduce nasal congestion. Binding of this drug with the alpha-1 receptor stimulates phospholipase C and increases IP3 and calcium as second messengers.

Question #: 12

A 34-year-old man comes to the physician because of a 2-week history of feeling lethargic and drowsy. Physical examination shows constricted pupils, which could result from overactive parasympathetic input. Stimulation of postganglionic parasympathetic neurons is most likely to cause which of the following additional effects?

A. Tachycardia
B. Smooth muscle vasodilation
✓C. Penile erection
D. Renin secretion
E. GI tract wall relaxation

Rationale: Point and Shoot
Point- parasympathetic -penile erection
Shoot- Sympathetic – ejaculation

Question #: 13

A 35-year-old woman comes to physician because of a 3-day history of worsening fever and malaise. Her pulse is 105/min, respirations are 18/min and blood pressure is 90/60 mmHg. She is diagnosed with septic shock secondary to an untreated bacterial infection. She is infused with a drug that increased vasoconstriction. Which of the following groups of drugs is most likely administered in this patient?

A. Alpha 1 adrenergic antagonists
B. Beta 1 adrenergic antagonists
✓C. Alpha 1 adrenergic agonists
D. Beta 2 adrenergic agonists
E. Muscarinic (M2) agonists

Rationale: The correct answer is c) Alpha 1 adrenergic agonists. In septic shock, the patient experiences a significant drop in blood pressure due to vasodilation and systemic inflammation. Alpha 1 adrenergic agonists are drugs that stimulate alpha 1 receptors, leading to vasoconstriction, which helps increase blood pressure. This makes them the most appropriate choice in this clinical scenario.
Drugs like phenylephrine or norepinephrine, which have alpha 1 agonist effects, are commonly used to manage hypotension in septic shock by constricting blood vessels and increasing systemic vascular resistance.
Alpha 1 adrenergic antagonists: These drugs block alpha 1 receptors, which leads to vasodilation, a decrease in blood pressure, and reduced vascular resistance.
Beta 1 antagonists block beta 1 receptors, primarily found in the heart. This leads to a decrease in heart rate and contractility, which can lower cardiac output and blood pressure.
Muscarinic M2 agonists stimulate the parasympathetic nervous system, leading to a decrease in heart rate and potentially lowering blood pressure.

Question #: 14

A 35-year-old man is brought to the Emergency Department after being severely injured in a car crash. His blood pressure is 50/30 mm Hg and respirations are 25/min. While attempting intubation for surgery, the doctor notices considerable airway secretions. What class of drug would reduce the amount of secretions in this patient?

A. Alpha 1-adrenoceptor agonist
B. Nonselective α-adrenoceptor antagonist
C. Beta 2-adrenoceptor antagonist
D. Muscarinic receptor agonist
✓E. Muscarinic receptor antagonist

Rationale: Muscarinic antagonists increase airflow in COPD by blocking cholinergic tone at airway smooth muscle. However, asthma is different in that airflow limitation is generally fully reversible and caused by bronchoconstriction. In more severe asthma, edema due to mucus hypersecretion also contributes to airflow limitation. The airways are hyperresponsive and bronchoconstrictor responses are exaggerated. Muscarinic antagonists increase airflow in asthma by blocking cholinergic tone and also by blocking reflex bronchoconstriction mediated by the vagus nerves. They may also inhibit secretion and clearance of mucus.

Question #: 15

A 50-year-old woman comes to the emergency department because of a 3-hour history of severe chest pain radiating to the left arm. Her pulse is 62/min, blood pressure is 140/86 mm Hg and respirations are 14/min. Physical examination shows profuse sweating, shortness of breath and dizziness. Electrocardiogram and cardiac markers are consistent with acute myocardial infarction. Which of the following best describes the sequence of events activating the sweat glands in this patient?

✓A. Postganglionic sympathetic fibers release acetylcholine, which binds to M3 receptors on the sweat glands to stimulate sweating
B. Postganglionic parasympathetic fibers release acetylcholine, which binds to M3 receptors on the sweat glands to stimulate sweating
C. Preganglionic sympathetic fibers release acetylcholine, which binds to M3 receptors on the sweat glands to stimulate sweating
D. Preganglionic parasympathetic fibers release acetylcholine, which binds to M3 receptors on the sweat glands to stimulate sweating
E. Postganglionic sympathetic fibers release norepinephrine, which binds to M3 receptors on the sweat glands to stimulate sweating

Rationale: The sympathetic nervous system innervates the sweat glands. In most of the cases, the postganglionic sympathetic fibers release norepinephrine. However, there is an exception to the rule: in the sweat glands, the sympathetic fibers release acetylcholine, which interacts with M3 receptors.

Question #: 16

An investigator is studying the effects of new compounds on arteriolar resistance. Drug X maximally increases vascular resistance by 50% at a dose of 20 mg/mL, and Drug Y maximally increases vascular resistance by 75% at 40 mg/mL. Which of the following best describes the results of the experiment?

A. Drug X has a smaller volume of distribution than Drug Y
B. Drug X has a shorter half-life than Drug Y

✓C. Drug X is less efficacious than Drug Y
D. Drug X is less potent than Drug Y
E. Drug X has a lower LD50 than Drug Y

Rationale: In this case the magnitude of the response (Efficacy) that drug Y produces is higher than the magnitude of Drug X. So overall Drug Y is more efficacious than Drug X or Drug X is less efficacious than Drug Y.

Question #: 17

A 15-year-old boy is brought to the emergency department by his parents because of agitation, mydriasis, hot, dry skin, and tachycardia. Assuming that he is suffering from a drug overdose, which of the following classes of drugs is most likely responsible for his symptoms?

✓A. Muscarinic receptor antagonist
B. Muscarinic receptor agonist
C. Acetylcholinesterase inhibitor
D. Beta adrenergic receptor antagonist
E. Beta adrenergic receptor agonist

Rationale: This is a classic example of poisoning with a muscarinic antagonist such as atropine or scopolamine. The anticholinergic effects are manifested such as mydriasis, agitation, tachycardia, decreased secretions which lead to increased Temperature due to inability to thermoregulate and dry skin.

Question #: 18

An investigator has developed a new drug. The new drug appears to be a partial agonist at receptor X. When the drug is applied to cells expressing receptor X, an influx of sodium is evoked. Receptor X is most likely which of the following?

A. Alpha1 adrenoreceptor
B. Beta1 adrenoreceptor
C. Beta2 adrenoreceptor
D. Muscarinic cholinergic receptor
✓E. Nicotinic cholinergic receptor

Rationale: This question tells us that the receptor is a ligand-gated ion channel. Out of the answer choices the nicotinic cholinergic receptors are ligand-gated ion channels. Beta receptors are coupled to Gs. Alpha 1 adrenoreceptors are coupled to Gq and Muscarinic receptors are coupled to Gq (M2 is coupled to Gi)

Question #: 19

A 25-year-old woman comes to the physician because of a 2-week history of altered vision. She had several episodes where she is beginning to "bump into things around the house". A CT-scan of her brain show areas that are devoid of myelin. Which of the following cells most likely normally produces myelin in this tissue?

A. Microglia
B. Astrocytes
✓C. Oligodendrocytes
D. Macrophages
E. Schwann cells

Rationale: The myelin sheath of peripheral nerves is produced by peripheral neuroglial cells called Schwann cells. The myelin sheath of the CNS is produced by central neuroglia called Oligodendrocytes.

Macrophages are phagocytic cells found primarily in the CNS. They scavenge cell debris.

Microglia are macrophages found in the CNS.
Astrocytes are neuroglial cells found in the CNS - they aid in the formation of the BBB.
Oligodendrocytes are myelin producing cells in the CNS

Question #: 20

A 52-year-old woman comes to the physician because of a 2-week history of pain and muscle weakness to both lower limbs. She had a severe gastrointestinal illness 4 weeks previously. Microscopic studies of the peripheral nerve fibers show an accumulation of lymphocytes, macrophages and plasma cells around nerve fibers and segments of the nerves are found to be devoid of myelin. Which of the following cells normally produce myelin in these nerves?

A. Oligodendrocytes
B. Microglia
✓C. Schwann cells
D. Astrocytes
E. Macrophages

Rationale: The myelin sheath of peripheral nerves is produced by peripheral neuroglial cells called Schwann cells.

Macrophages are phagocytic cells found primarily in the CNS. They scavenge cell debris.

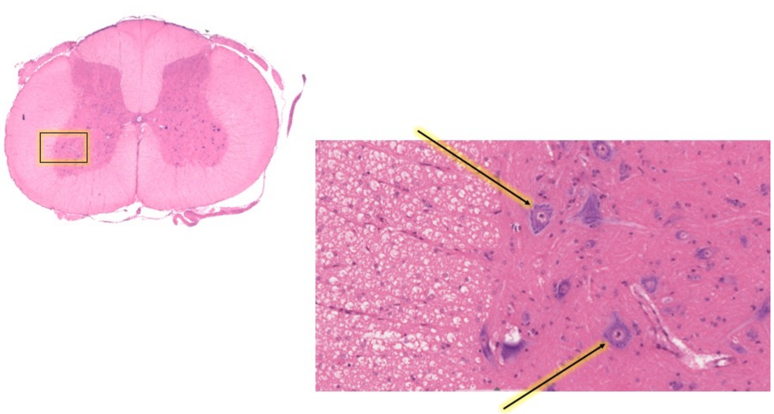
Microglia are macrophages found in the CNS.
Astrocytes are neuroglial cells found in the CNS - they aid in the formation of the BBB.
Oligodendrocytes are myelin producing cells but in the CNS

Question #: 21

A 45-year-old man is brought to the emergency department because of back pain following a fall from his roof. Physical examination shows a loss of movement to the lower limbs. An x-ray of the spine shows a crush injury and fractures to the lumbar level of the vertebral column. An MRI shows the spinal cord to be intact but significantly swollen, characteristic of spinal shock. An image of the spinal cord is attached. Which of the following cells are characteristic of the highlighted region?

A. Sensory interneurons
B. Purkinje cells
C. Pyramidal cells
✓D. Alpha motor neurons
E. Ependymal cells

Attachment:



VentralHorn.jpg

Question #: 22

A research group is investigating the side effects of a new antibiotic on the brain. The drug is administered to rodents over a period of 2 weeks Laboratory studies show that the drug or its metabolites are present in blood but not in the cerebrospinal fluid. Biopsy studies show that the drug was not present in the brain tissue. Which of the following cells is most likely a component of the structure which prevented the drug from entering the brain tissue?

A. Schwann
B. Ependymal
C. Oligodendrocyte
✓D. Astrocytes
E. Microglia

Rationale: The blood brain barrier regulates passage of substances between the endothelium of the cerebral vessels and the brain tissue. Astrocytes are a major component of this structure.

Question #: 23

A 32-year-old woman comes to the physician because of a 6-month history of lower abdominal pain during her menstrual periods. She has heavy menstrual bleeding during every cycle and intense cramping pain. Physical examination shows no abnormalities. She is diagnosed with menstrual cramps due to the contraction of the smooth muscle of the uterus. Which of the following structures are associated with Ca^{+} delivery to the cytoplasm of these muscle cells?

A. Condensation bodies
B. Dense bodies
C. T-tubules
✓D. Caveolae
E. Intercalated discs

Rationale: Caveolae are invaginations of the plasma membrane of smooth muscle cells that allows the membrane to be in much closer contact with Ca2+ ions, which are found in the sER.

Dense bodies form the attachment site for the contractile proteins, as well as the intermediate filaments found in smooth muscle cells (type III intermediate filaments). They are usually located both on the cell's membrane, as well as within the sarcoplasm.

T-tubules are not a feature of smooth muscle cells, but are found in striated muscles. The caveolae of smooth muscle is analagous to the T-tubules of striated muscles.

Intercalated discs are not found in smooth muscle tissue, but rather in cardiac muscle tissue.

Question #: 24

A 37-year-old man is brought to the emergency department because of a 4-hour history of sudden, severe chest pain. He dies despite appropriate medical care. Autopsy shows a severe form of myocarditis (inflammation of the cardiac muscle). Which of the following best characterizes this type of muscle tissue?

A. Forms sheets in the wall of hollow organs
✓B. Composed of branched, mono or bi-nucleated cells
C. Composed of long, cylindrical, multi-nucleated cells
D. Composed of fusiform, mono-nucleated cells
E. Forms the muscles of the body wall

Rationale: Cardiac myocytes are branched, and they may have one or two centrally-placed nuclei

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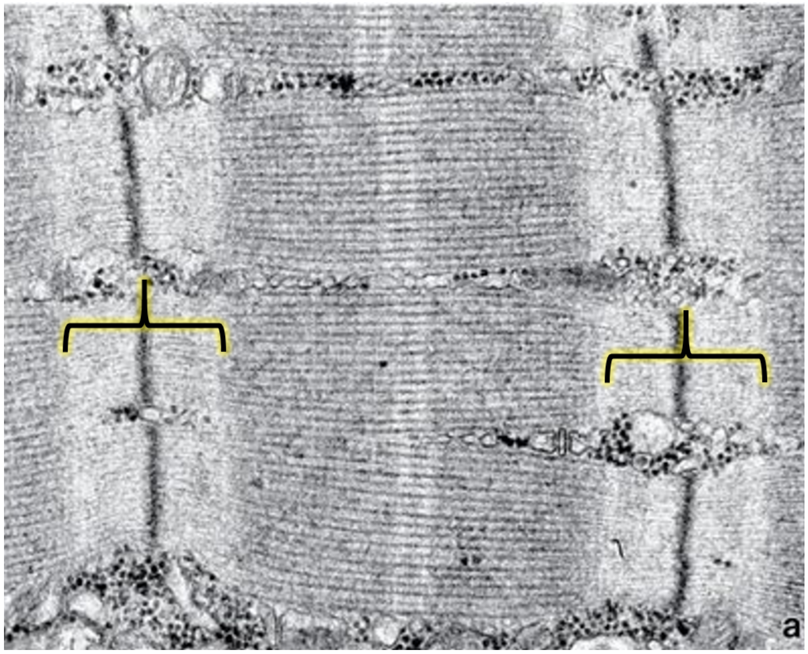
Question #: 25

A 35-year-old man is brought to the emergency department because of recurrent seizures. Physical examination shows tonic-clonic seizure with spasm of the skeletal muscles. Which of the following best characterizes the structure indicated by the brackets on the attached electron micrograph?

A. Presence of thick filaments only
✓B. Presence of thin filaments only
C. Presence of thick and thin filaments
D. Absence of any kind of filaments
E. Makes up a sarcomere

Rationale: The I band is characterized as an area where there are thin filaments only.
Attachment:



I band Muscle.png

Question #: 26

A 45-year-old man is brought to the emergency department because of sudden, severe chest pain. He dies despite appropriate medical care. Autopsy shows a massive myocardial infarction. Which of the following best characterizes the location of the diad in the type of muscle tissue most likely affected?

A. At the A-I junction
✓B. At the Z-line
C. In the A-band
D. Center of H-band
E. Within the M-line

Rationale: The sarcoplasmic reticulum and T tubule form a diad at the level of the Z line in cardiac muscle.

Question #: 27

A 20-year-old man comes to the physician because of a 1-week history of a painful skin rash. He received a bee sting 2 weeks ago. Physical examination of the affected area shows redness and mild swelling. A biopsy is taken from the rash and the slides are studied using different stains. One of them was toluidine blue, which brings out the property of metachromasia within certain cells of the connective tissue. Which of the following cells are most likely responsible for this reaction?

✓A. Mast cells
B. Fibroblasts
C. Macrophages
D. Neutrophils
E. Lymphocytes

Rationale: Mast cells are found in connective tissue. They are very similar to basophils in composition and function. Their cytoplasmic granules contains histamine and heparin. These granules can bind very well to hematoxylin, but when stained with toludine blue, the color of the granules change to red. This property is called metachromasia. None of the other cells listed carries this property.

Question #: 28

A 35-year-old man is admitted to the hospital because of a 10-day history of high fever, chest pain and cough. Physical examination shows a febrile patient with decreased breath sounds on both sides of his chest. An x-ray shows bilateral pneumonia of his lungs and a heart that is located on the right side. A diagnosis of Kartagener's syndrome is made. Which of the following motor proteins are most likely affected in this condition leading to these symptoms?

A. Actin
B. Spectrin
✓C. Dynein
D. Myosin
E. Kinesin

Rationale: This syndrome usually affects the cilia. The axoneme of the cilia uses the motor protein dyenin for movement of the cilia.

Actin is for contraction, along with myosin to promote cell motility eg fibroblasts, macrophages or to promote changes in plasma membrane for endocytosis.

Kinesin is a motor protein involved in intracellular movement.
Spectrin is a protein found in the membrane of RBCs to maintain the integrity of that membrane.

Question #: 29

A 72-year-old man is brought to the physician because of a 3-month history of short and long term memory loss. He is admitted for investigations but soon dies in the hospital. An autopsy is performed to identify the cause of his memory loss and the findings are shown in the attached photomicrograph. Which of the following locations is this biopsy most likely taken from?

✓A. Cerebral cortex
B. Cerebellum
C. Dorsal root ganglion
D. Spinal cord
E. Sympathetic ganglion

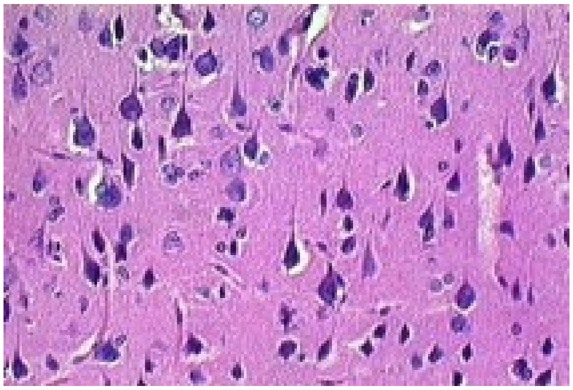
Rationale: The presence of the pyramidal cells are characteristic of the cerebrum. The cell bodies of the motor neurons found in the gray (outer) white matter of the brain.

The gray matter of the cerebellum has 3 layers. The purkinje cells are prominent in this tissue.

In the spinal cord, the gray matter with the cell bodies are found on the inner layer.

Ganglia are arranged differently.

Attachment:



Cerebrum.jpg

Question #: 30

A 34-year-old woman comes to the physician because of a 5-day history of a swelling on her right thigh. Physical examination shows a mass within the extensor muscle compartment of the thigh. A biopsy is obtained and shown in the attached micrograph. Which of the following images best represents the biopsied muscle type?

A. 1
✓B. 2
C. 3
D. 4
E. 5

Rationale: Limb muscles are examples of skeletal muscle, which is characterized by bundles/fascicles of striated, long, multinucleated muscle fibers with peripherally located nuclei (Option 2).

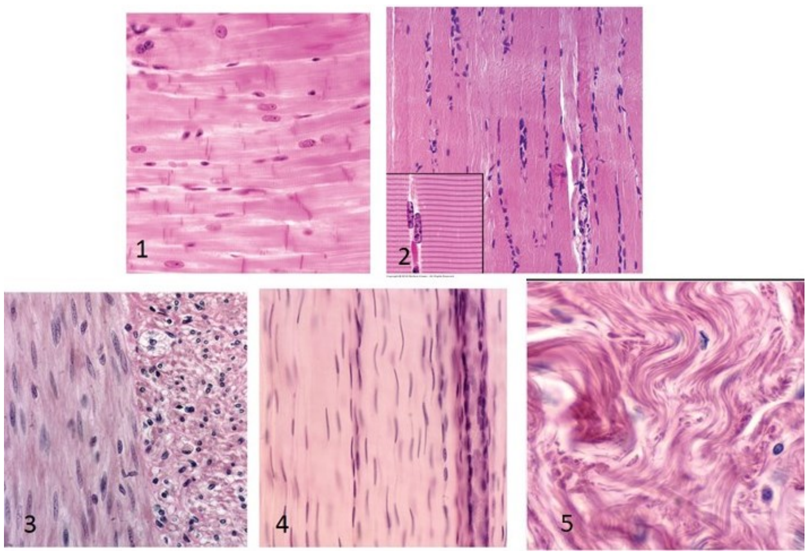
Option 1- Cardiac muscle (stiated, branched, central nucleus or binucleated, intercalated discs).

Option 3- Smooth muscle (no striations, bundles or sheets oriented in different directions, fusiform cells with central nucleus).

Option 4- Dense Regular CT (more fobers than cells, fibers in parallel array and densely packed).

Option 5- Dense irregular CT (more fibers than cells, irregular arrangement of fibers)

Attachment:



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